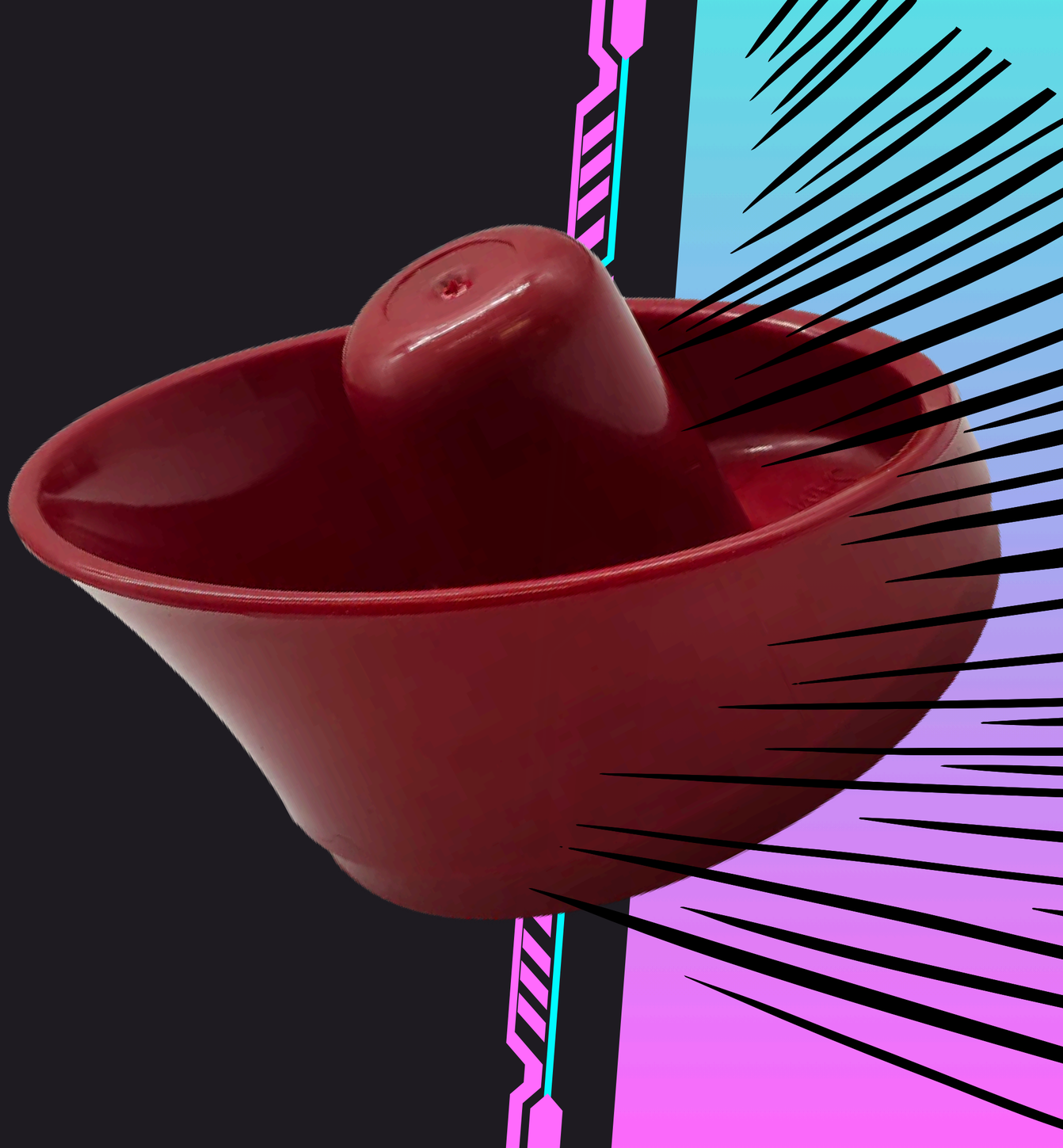
JACOB MARTINEZ

AIR HOCKEY ARENA



WELCOME TO THE RINK

A NEW SPIN ON AIR HOCKEY

Air Hockey is a Classic Arcade Game that I often have in my mind.
When I saw the pushblock homework I was immediately inspired to push other things, like a hockey puck.
Not too much innovation on the Airhockey side.
I thought it would be fun to modify the field with certain obstacles or powerups.

GENERAL IMPLEMENTATION

EXPORTED FROM ASSIGNMENT 4 PUSH BLOCK
MODELED AFTER THE SOCCAR STRUCTURE

There are two mallets, either controlled by players or agents that can hit a frictionless and bouncy puck towards a goal area, which causes a collider trigger that resets the stage.

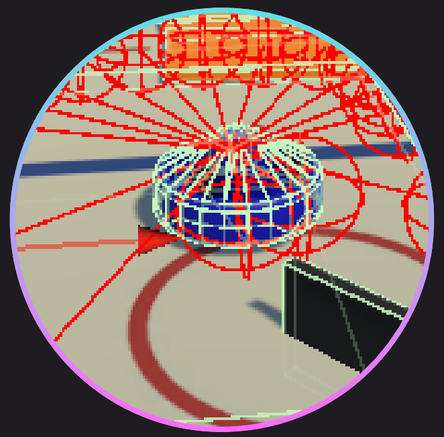
The only thing that is moveable on the stage is the puck. There are invisible colliders that give the puck its bouncing feel.

If you are playing, the controls are WASD and you are only allowed to move on your side of the rink.

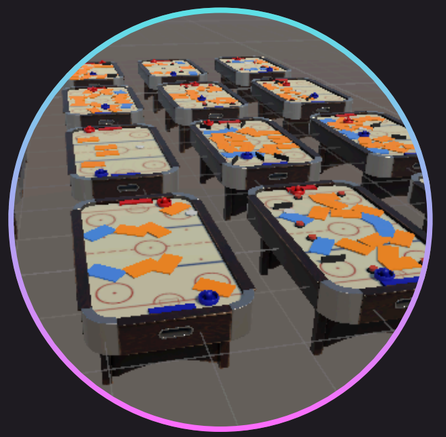
4 BIG TICKET ITEMS

Perlin Noise

New
Enviornments



Agent Action
Space



Training
Regimine

PERLIN NOISE

SEEDED PERLIN MAPS + ADDITION OFSET MODIFIERS

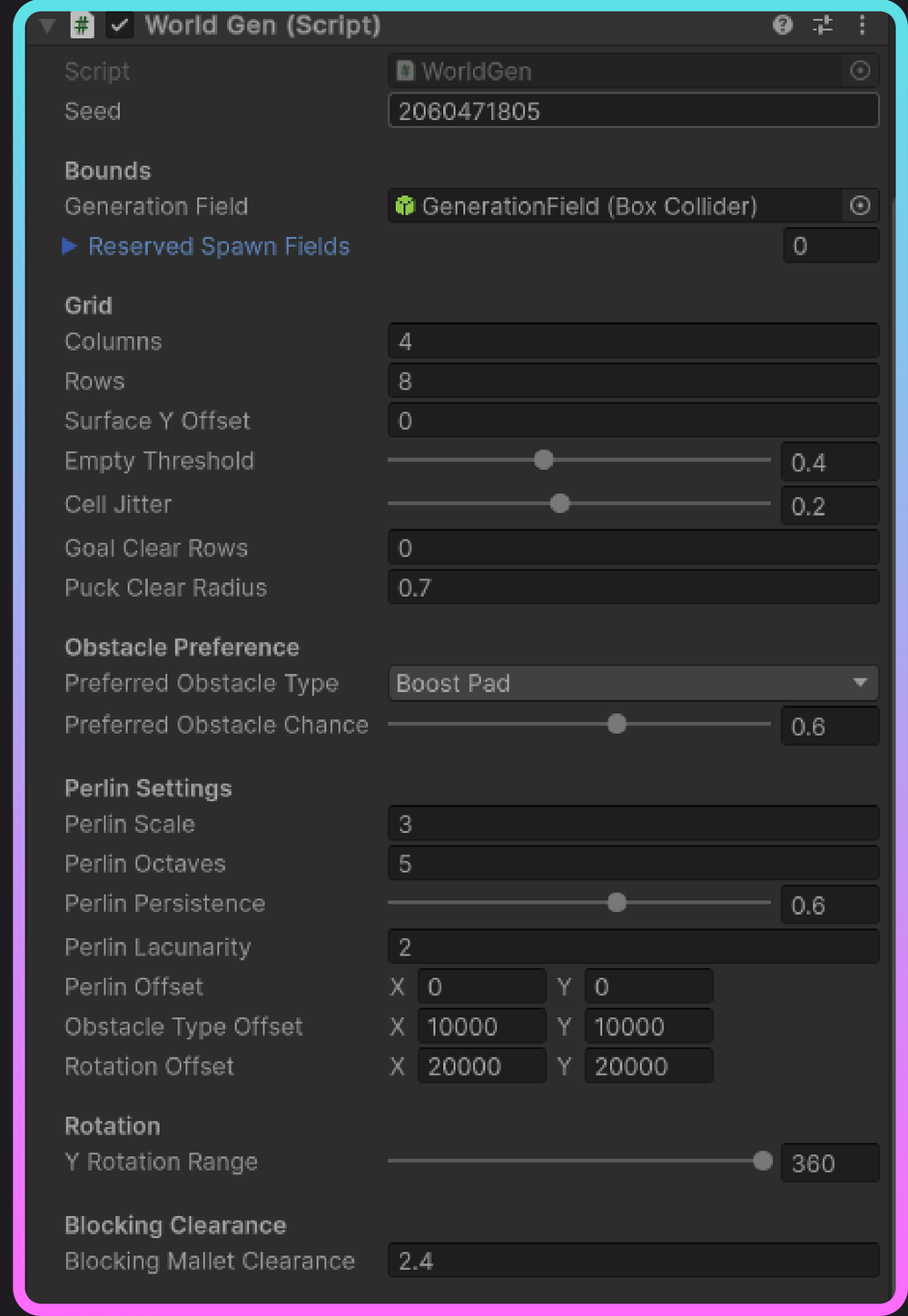
The Perlin Noise Map is read 2 different Ways:

1. Determines what object is playced onto the field grid.
2. Determines the orientation of the object.

Has a low scale so we see a variety of different objects within each stage, a medium amount of octaves and persistence to balance the detail and a low lacunarity to keep a smooth balance of objects.

You can adjust the likelyhood of an object appearing under obstacle preference.

Allowing for more “themed” maps



PERLIN NOISE

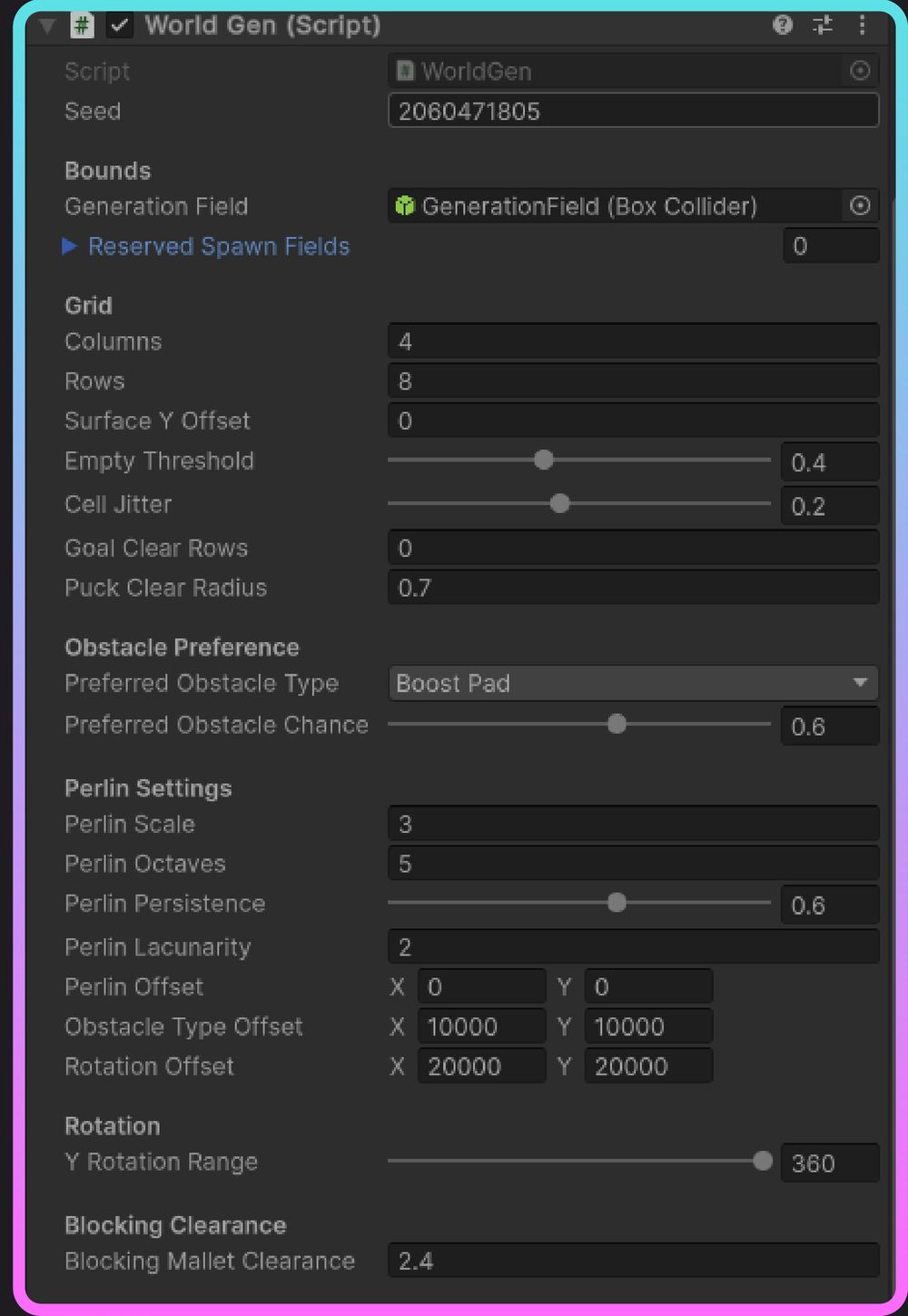
SEEDED PERLIN MAPS + ADDITION OFSET MODIFIERS

The World Gen breaks a continous space into specified grid sections

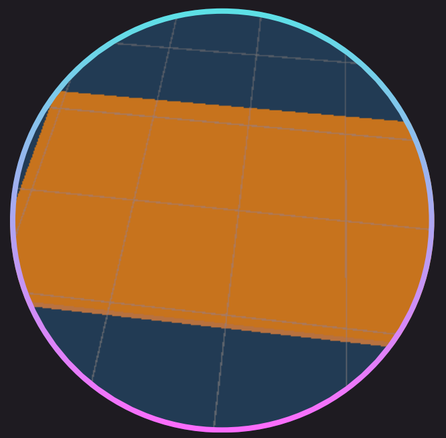
Safe guards prevent objects from obstructing the player by removing them from the grid before they are spawed.

This specifically clears space in front of the goal, around the puck starting area, and removes objects that would block the mallet's clearence.

The mallet checks for an empty area before it spawns in.

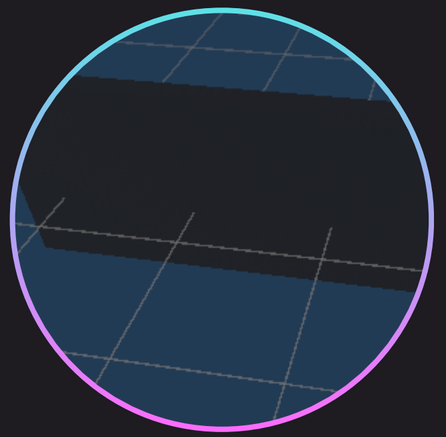


4 ENVIORNMENT MODIFIERS

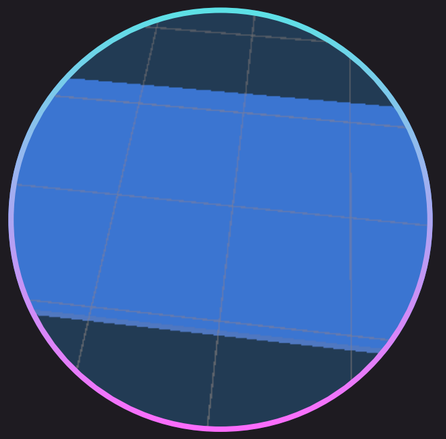


Boost Pad

Bumper



Wall



Slow Zone

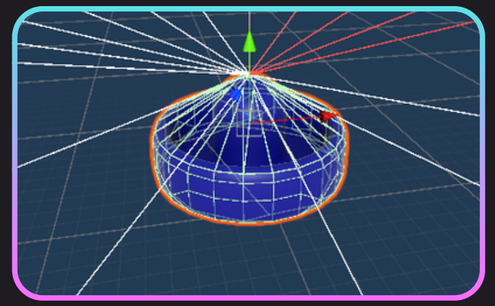
MLAGENT DESIGN - ACTION SPACE

INTERPRETS THE SOUNDINGS WITH 2 SETS OF RAY SENSOR 3DS

Information is found by the ray cast and sent to collect observations.

IIncluding walls, obstacles, other mallets and the puck.

Includes objects it sees and its own relative position and velocity.



```
lic override void CollectObservations(VectorSensor sensor)

if (sensor == null)
{
    return;
}

if (m_EnvController == null)
{
    m_EnvController = GetComponentInParent<AirHockeyEnvController>();
}

AddVectorObservation(sensor, transform.position - CenterPosition);
AddVectorObservation(sensor, agentRb != null ? agentRb.velocity : Vector3.zero);
AddVectorObservation(sensor, m_Puck != null ? m_Puck.position - transform.position : Vector3.zero);
AddVectorObservation(sensor, m_PuckRb != null ? m_PuckRb.velocity : Vector3.zero);
AddVectorObservation(sensor, m_Opponent != null ? m_Opponent.position - transform.position : Vector3.zero);
AddVectorObservation(sensor, m_OpponentRb != null ? m_OpponentRb.velocity : Vector3.zero);
```

```
var discreteActionsOut = actionsOut.DiscreteActions;
discreteActionsOut[0] = 0;
discreteActionsOut[1] = 0;

if (Input.GetKey(KeyCode.W))
{
    discreteActionsOut[0] = 1;
}
else if (Input.GetKey(KeyCode.S))
{
    discreteActionsOut[0] = 2;
}

if (Input.GetKey(KeyCode.D))
{
    discreteActionsOut[1] = 1;
}
else if (Input.GetKey(KeyCode.A))
{
    discreteActionsOut[1] = 2;
```

Two Discrete Action Branches:

- Forward/Back/None
- Left/Right/None

```
void RewardPuckHit(Rigidbody puckRb, Vector3 preHitPuckVelocity, Vector3 strikeImpulse)
{
    Academy.Instance.StatsRecorder.Add("AirHockey/PuckHits", 1f);

    var rewardMultiplier = GetPuckHitRewardMultiplier();
    m_PuckHitsThisEpisode += 1;
    if (rewardMultiplier <= 0f)
    {
        return;
    }

    var expectedLaunchVelocity = preHitPuckVelocity;
    if (puckRb != null)
    {
        expectedLaunchVelocity += strikeImpulse / Mathf.Max(puckRb.mass, 0.0001f);
    }

    expectedLaunchVelocity.y = 0f;
    var launchSpeed = Mathf.Min(expectedLaunchVelocity.magnitude, MaxPuckSpeed);
    var velocityBonus = Mathf.Clamp01(launchSpeed / PuckHitVelocityForMaxBonus) * MaxPuckHitVelocityBonus;
    var hitReward = (PuckHitReward + velocityBonus) * rewardMultiplier;
```

MLAGENT DESIGN - REWARDS

FOCUSING ON FAST HITS AND FIRST HITS

Rewards terminal conditions and penalizes timeout lightly.
Has additional reward that decreases within the first 5 hits, then is no longer rewarded.
These puck hits also have a velocity bonus.

TRAINING REGIMINE

MULTISTAGE CURRICULUM

- Behavioral Cloning 500k
- Self Play (Basic Field) 3.5M
- Self Play (Random Enviornment) 1M
- Self Play (Specialized Enviornment) 500k

SIMULATES MULTIPLE DIFFICULTY LEVELS

```yaml
trainer_type: ppo
hyperparameters:
  batch_size: 2048
  buffer_size: 20480
  learning_rate: 0.0003
  beta: 0.005
  epsilon: 0.2
  lambd: 0.95
  num_epoch: 3
  learning_rate_schedule: constant
network_settings:
  normalize: false
  hidden_units: 256
  num_layers: 2
  vis_encode_type: simple
reward_signals:
  extrinsic:
    gamma: 0.99
    strength: 1.0
checkpoint_interval: 1000000
keep_checkpoints: 5
max_steps: 3500000
time_horizon: 1000
summary_freq: 10000
self_play:
  save_steps: 50000
  team_change: 200000
  swap_steps: 2000
  window: 10
  play_against_latest_model_ratio: 0.5
```

PLAYTESTS

GOAL PATHWAY TEST

Generates random board, finds a path from the center to both goal and checks visible path segements.

The test fails if there is no clearence for the puck to reach the goal.

This is to ensure that generation always has the opportunity to score.

MALLET FREEDOM TEST

Mallet freedom test flood-fills open space and requires at least 80% reachability.

This ensures that there is enough playspace for the player and agents to hit the puck from various angles.

ULTIMATELY THESE ALSO PREVENT SEVERE OVERCROWDING

CHALLENGES

WORLD GEN ON A CONTINOUS SPACE

My biggest problem was taking the tile based gen from assignment 3 and converting it.
Bounced between two options:

- Continous genration, but with bounded clearence on each object
- Grid based segementation, plus offset

Causes many a crash

HOW WOULD I PROCEDE IN THE FUTURE

ADD ROGUE-LIKE ELEMENTS

- Progression through a gauntlet, you get to customize the field (choose between options)
- Adding that same choice to the agent's action space.
- Include a difficiculty rating for each level after it is generated.
- Add more player features.

DEMO

https://lmu.box.com/s/onj5rxc92mj5yqtolgc6lddidaqtdk5